



Advancing environmental toxicology through sustainable technologies, intelligent systems, and forensic innovations

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Environmental toxicology stands at the intersection of research, sustainability, and societal imperative. Environmental danger is rapidly altering due to industrialization, climate change, and emerging poisons. This Special Issue is based on the compilation “Advancements in Environmental Toxicology: Assessing Human Exposure and Health Risks” and comprises a range of interconnected studies that tackle pressing concerns in environmental contamination, monitoring, remediation, and sustainable technological progress. It emphasizes integrated approaches for risk evaluation and environmental surveillance. Collectively, these contributions provide critical insights for advancing research and policy in environmental toxicity.

A prominent subject in this collection is the shift towards sustainable resource utilization and eco-friendly technical solutions. Research on the thermal pretreatment of water hyacinth suggests the feasibility of utilizing invasive biomass to produce value-added biofuels, including bioethanol and biomethanol. It exemplifies a circular economy by integrating waste management with renewable energy generation. Furthermore, advancements in sustainable machining through nanofluid-assisted minimum quantity lubrication demonstrate how technological innovations can significantly reduce environmental impact without compromising industrial output.

Environmental toxicology remains concentrated on the management of water resources and the monitoring of environmental conditions. The application of GIS technologies and multi-criteria evaluation for assessing groundwater potential in the watershed provides a solid foundation for sustainable water resource planning. The examination of land cover alteration and the impact of slope gradient on soil physicochemical properties yields critical insights into the effects of anthropogenic and geomorphological influences on soil health and contaminant dynamics. Another significant component of this collection is the domain of developing contaminants and novel materials. A comprehensive evaluation of lithium in the Anthropocene underscores remediation strategies while also expressing apprehensions regarding the escalating accumulation, bioavailability, and health ramifications of lithium in the environment. The synthesis of MWCNT including Cu-BDC metal–organic frameworks for the photocatalytic degradation of methylene blue dye demonstrates the potential of nanostructured materials in addressing industrial wastewater pollution. Artificial intelligence and data-centric methodologies are revolutionizing environmental toxicology. The utilization of explainable deep reinforcement learning in climate forecasting provides a transparent and adaptive modeling framework, while AI/ML-driven computational models for toxicity prediction signify a transformative shift towards predictive toxicology, departing from conventional experimental techniques and facilitating expedited risk assessment. The compilation is a notably innovative contribution to the domain of green forensics. This study presents the effective use of an eco-friendly eggshell-derived reagent for the detection of latent fingerprints on objects immersed in water, highlighting the potential of sustainable materials in forensic science and bridging green chemistry with forensic investigation.

These contributions together present a multidisciplinary strategy that integrates environmental chemistry, engineering, data science, and forensic methodologies. The research concentrates on the detection and analysis of pollutants, as

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well as sustainable remediation, risk mitigation, and policy-relevant findings. The collection emphasizes the significance of integrative strategies employing green technology, advanced materials, and intelligent systems to address complex environmental toxicological issues. It underscores the necessity of progressing from laboratory-scale research to field validation, scaling, and practical implementation. This pertinent subject can provide valuable insights for scholars, policymakers, and practitioners involved in sustainable environmental management and pollution mitigation. It also underscores the promotion of future multidisciplinary research that enhances environmental safety and resilience.

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Declarations

Conflict of interest None

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Kumud Kant Awasthi is a Professor of Life Sciences with more than 16 years of experience in teaching, research, and academic leadership. He holds a Ph.D. in Zoology and specializes in environmental toxicology, nanotoxicology, and bioremediation. His research focuses on the biological effects of nanoparticles, heavy metal contamination, oxidative stress mechanisms, and sustainable strategies for soil and water remediation. He also works on nano-enabled biosensors and forensic applications of nanotechnology. Dr. Awasthi has published over 125 peer-reviewed research articles in reputed international journals and has edited multiple academic books with leading publishers such as Springer, Elsevier, Wiley, and CRC Press. He holds several patents related to nanotechnology and environmental applications. His work integrates fundamental biological research with applied environmental solutions. He has supervised numerous postgraduate and doctoral research scholars and has played a key role in establishing academic programs and laboratories in life sciences and forensic science. He actively contributes to the scientific community as an editor, guest editor, and reviewer for international journals. His research aims to address environmental challenges through innovative, sustainable, and interdisciplinary scientific approaches.

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Mahipal Singh Sankhla is an environmental scientist and researcher with expertise in environmental forensics, environmental pollution, toxicology, and sustainable remediation strategies. His research focuses on heavy metal contamination, water and soil pollution, environmental health, and the development of eco-friendly solutions for waste management and resource recovery. He has contributed significantly to understanding the impact of pollutants on ecosystems and human

health, with a strong emphasis on practical and sustainable approaches. Dr. Sankhla has published numerous peer-reviewed research articles, review papers, and book chapters in reputed international journals and publishers. His work includes studies on heavy metal toxicity, wastewater treatment, climate-related environmental changes, and circular economy approaches. He is also actively involved in interdisciplinary research integrating environmental science with materials science and biotechnology. He has experience in research supervision, academic mentoring, and collaborative projects, contributing to the development of young researchers. Dr. Sankhla is also engaged in editorial and peer-review activities for international journals. His research aims to bridge scientific knowledge with real-world environmental challenges, focusing on sustainable development, pollution control, and innovative environmental technologies.



Rajeev Kumar is a Professor and Head of the Department of Forensic Science, IILM University, Greater Noida, UP, India, with over a decade of experience in teaching, research, and academic leadership. He holds a Ph.D. in Forensic Science with research focused on heavy metal contamination and toxicity in aquatic systems, particularly the Yamuna River. His expertise spans forensic toxicology, environmental pollution, analytical techniques, and nanotechnology-based forensic applications. He has published

more than 110 research papers, authored book chapters, edited books, and holds multiple patents, including granted and published innovations in forensic and sustainable materials. His research contributions have received over 2200 citations with a strong h-index, reflecting significant academic impact. Dr. Kumar has extensive experience in research supervision, having guided Ph.D. and postgraduate students, and has served in key academic roles including Program Chair and Head of Department. He is actively involved in editorial boards of reputed journals and has been recognized with awards such as the Young Academician Award. His work integrates forensic science, environmental monitoring, and applied research to address real-world challenges.